Product Requirement Document (PRD): Edge Drone Vision Surveillance

AUTONOMOUS UAV REAL-TIME VISION & COLLISION AVOIDANCE SPECIFICATION

1. Project Executive Overview

This specification outlines hardware requirements for deploying an onboard real-time computer vision system for agricultural and perimeter surveillance UAVs in India.

2. Technical & Hardware Constraints Matrix

Parameter	Specification Value	Architectural Note
Model Architecture	MobileNetV2 / YOLOv8 Nano Edge Vision	Lightweight inverted residual convolutional network
Model Parameter Count	25 Million parameters (0.025B)	Target footprint under 50 MB in memory
Weight Precision	INT8 Quantized (PTQ / QAT)	Optimized for integer tensor processing cores
Target Output Time / Latency	18 ms per frame (under 20 ms)	Hard real-time deadline for autonomous collision avoidance
Target Throughput	45 FPS (Frames Per Second)	Continuous video stream processing at 1080p resolution
Target Budget	INR 25,000 (Rs. 25,000)	Unit bill-of-materials capital expenditure ceiling
Thermal Power Envelope	15 Watts TDP (under 18 W)	Strict battery-powered payload limitation for UAV flight endurance
Batch Size	1 frame	Zero buffering for instantaneous closed-loop steering
Sovereign Silicon Preference	Make in India / High Preference	Preference for domestic Indian processors such as Vicharak Vaaman

3. Deployment & Operational Requirements

The drone payload system requires continuous edge vision inference deployed on an onboard single-board computer (SBC). To maximize battery endurance, total system power consumption must remain strictly below 15 Watts TDP. The camera feed is processed at 45 FPS with an output latency barrier under 18 ms. Cost efficiency is paramount; total accelerator budget is capped at INR 25,000 (Rs. 25,000) per unit. Indian sovereign hardware like the Vicharak Vaaman (Rockchip RK3588 NPU) or Jetson Orin Nano is evaluated for domestic deployment.